Intermediate Mathematics: Syllabus

Jacob Khohayting

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※ Methodology

Each weekly, one-hour session combines clear instruction, collaborative problem-solving, and opportunities to explore ideas beyond the standard curriculum.

Our classes emphasize reasoning, communication, and discovery. Students will learn useful mathematical techniques, but they will also investigate why those techniques work, when they are most effective, and how to communicate their thinking clearly.

Sessions are designed to be welcoming and interactive. Students will strengthen their mathematical foundations, ask ambitious questions, and encounter new ideas alongside their peers. Each course offers a short, focused opportunity to build confidence, develop problem-solving skills, and take on a fresh mathematical challenge.

All classes are offered completely free of charge as part of our commitment to making mathematical enrichment accessible to students in our community.

※ Prerequisites

Students should be comfortable working with the following topics in our introductory class:

- Algebraic Concepts including, but not limited to solving systems of equations, word and rate problems, and knowledge of mean, median, mode, and range.
- Number Theory Concepts including, but not limited to divisibility, primes, and prime factorization.
- Combinatorial Concepts including, but not limited to factorials, combinations, and permutations.
- Geometric Concepts including, but not limited to area and perimeter.

Students are encouraged to complete our

20-problem readiness assessment

before registering. A student is likely prepared for the course if they can solve a majority of the problems or make substantial progress on many of them. Students are not expected to solve every problem perfectly. In addition, students may still participate in the introductory class.

The assessment is intended as a guide rather than a formal entrance examination. For questions about placement or readiness, please contact us using the information in the *Further Questions?* section.

※ About the Instructors

Jacob Khohayting

Jacob began participating in mathematics competitions in elementary school. He qualified for the 2025 USA Junior Mathematical Olympiad (USAJMO), where he earned an Honorable Mention. His favorite area of mathematics is geometry, and he has earned top-ten finishes at national mathematics competitions hosted by Carnegie Mellon University, Harvard University, the Massachusetts Institute of Technology, and Princeton University.

Joshua Khohayting

Joshua has qualified for the American Invitational Mathematics Examination (AIME) four times since seventh grade. Beyond competition mathematics, he has pursued advanced mathematical study through the 2025 Hampshire College Summer Studies in Mathematics.

Students are encouraged to reach out to the instructors with mathematical questions. They may also submit written solutions for individualized feedback on their reasoning, clarity, and proof-writing.

※ Topics

Courses may draw from the following areas, depending on student readiness and the focus of each session:

Absolute Value and Inequalities: Understanding Distance and Bounds Interpret absolute value as distance, solve equations and inequalities, and represent solution sets algebraically and visually on the number line.

Geometric Reasoning: Right Triangles and Angle Chasing Apply the Pythagorean Theorem, use fundamental angle relationships, and develop organized strategies for multi-step angle-chasing problems.

Beyond Triangles: Circles and Polygons Explore the properties of circles, quadrilaterals, and other polygons, including angle relationships, side relationships, and useful geometric constructions.

Sequences and Series: Patterns, Sums, and Telescoping Investigate arithmetic and other patterned sequences, derive formulas for their terms and sums, and simplify series through telescoping and creative pattern recognition.

Modular Arithmetic: Remainders and Legendre's Formula Develop fluency with modular arithmetic and congruences, then apply Legendre's formula to study prime factors and divisibility within factorials.

Recursive Thinking: Building from Previous Terms Define sequences and processes recursively, identify patterns generated by recurrence relations, and use earlier information to determine later behavior.

Polynomial Problem-Solving: Factoring and Vieta's Formulas Strengthen polynomial manipulation and factoring skills while using Vieta's formulas to connect the roots of a polynomial with its coefficients.

Strategic Counting: Complements, Cases, and Arrangements Solve combinatorial problems through complementary counting, systematic casework, and careful analysis of arrangements involving restrictions or repeated objects.

The goal is not merely to cover a list of topics, but to help students recognize connections among them and apply their knowledge creatively.

※ Interested?

Registration is available through our

online registration form.

There is no tuition or registration fee. Every class is taught by accomplished student mathematicians who are excited to share their experience, curiosity, and enthusiasm for problem-solving.

※ Further Questions?

For questions about registration, course placement, readiness, or class content, please contact us at

jacobkhohayting@gmail.com.

※ Course Information

Format: Live online instruction.

Session Length: One hour.

Meeting Schedule: Monday, Wednesday, and Friday at 1 PM – 2 PM.

Time Zone: Eastern Time Zone.

Online Platform: Zoom. Links will be sent upon registration.

Intended Students: 3rd to 5th graders, but all are welcome.

Class Size: Large. However, students are welcome to ask questions and email instructors for further support.

Cost: Free of charge.

※ Learning Objectives

By participating actively in the course, students will work toward the ability to:

- approach unfamiliar problems with confidence and persistence;
- select appropriate strategies rather than relying on memorized procedures alone;
- identify patterns and make useful mathematical conjectures;
- explain solutions clearly in both written and spoken form;
- evaluate and improve their own reasoning;
- collaborate respectfully with other students; and
- recognize connections among arithmetic, algebra, geometry, number theory, statistics, and combinatorics.